

Reward-Tilted Posterior Guidance for Discrete Diffusion

Experimental log: QM9 property guidance

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1 Method

The reverse posterior marginalizes over the clean sequence,

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t) = \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0 \mid \mathbf{x}_t)} [q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0)], \quad (1)$$

which we importance-sample using the model’s denoiser $p_\theta(\mathbf{x}_0 \mid \mathbf{x}_t)$ as the proposal. Because the forward noising process is shared between q and p_θ ,

$$\frac{q(\mathbf{x}_0 \mid \mathbf{x}_t)}{p_\theta(\mathbf{x}_0 \mid \mathbf{x}_t)} = \frac{q(\mathbf{x}_0)}{p_\theta(\mathbf{x}_0)} \cdot \frac{p_\theta(\mathbf{x}_t)}{q(\mathbf{x}_t)}, \quad (2)$$

and the second factor does not depend on $\mathbf{x}_0$, so it cancels under normalization. Defining the reward-tilted target $q(\mathbf{x}_0) \propto p_\theta(\mathbf{x}_0) \exp(\lambda r(\mathbf{x}_0))$ leaves the importance ratio equal to $\exp(\lambda r(\mathbf{x}_0))$, giving

$$q(\mathbf{x}_{t-1} \mid \mathbf{x}_t) \approx \sum_{n=1}^N w_n q(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0^{(n)}), \quad w_n = \frac{\exp(\lambda r(\mathbf{x}_0^{(n)}))}{\sum_{m=1}^N \exp(\lambda r(\mathbf{x}_0^{(m)}))}, \quad (3)$$

with $\mathbf{x}_0^{(n)} \sim p_\theta(\mathbf{x}_0 \mid \mathbf{x}_t)$.

Relation to discriminator guidance. Discriminator guidance fills the same slot with a *learned* density ratio, using $D_\phi(\mathbf{x}_0)/(1 - D_\phi(\mathbf{x}_0)) \approx q(\mathbf{x}_0)/p_\theta(\mathbf{x}_0)$ from a discriminator trained to separate data from model samples. Our method obtains the same ratio in closed form by *defining* the target as a reward tilt, so nothing has to be trained. This is also the key practical difference from D-CBG: the reward sees *clean* $\mathbf{x}_0$ samples, so it never needs to be robust to noised inputs, and each denoising step costs one network forward (the same as unguided sampling) plus N CPU-side RDKit evaluations.

Sampling the mixture. Equation (3) is a mixture over *full sequences*, and a mixture of factorized distributions is not itself factorized. Three variants are implemented, forming a ladder of approximations:

- **exact** — draw $n^* \sim \text{Cat}(w)$, then $\mathbf{x}_{t-1} \sim q(\cdot \mid \mathbf{x}_t, \mathbf{x}_0^{(n^*)})$. Samples Eq. (3) without further approximation and preserves correlations across positions.
- **marginal** — average the N *normalized* posteriors and sample positions independently. Exact per-position marginals, no correlations.

- **aggregate_x0** — substitute the weighted histogram $\bar{\mathbf{x}}_0$ of the N candidates for $\mathbf{x}_0$ in the posterior and normalize once. This is the form discriminator guidance uses; it coincides with **marginal** for absorbing-state diffusion, whose posterior normalizer does not depend on $\mathbf{x}_0$, and differs slightly for uniform diffusion.

Note that **aggregate_x0** reduces to the base sampler as $N \rightarrow \infty$ at $\lambda = 0$, since $\bar{\mathbf{x}}_0$ is then an unbiased Monte Carlo estimate of $\mathbf{x}_\theta$. **exact** does not, at any N — but that is because the base sampler is itself the factorized projection of the true reverse step, which **exact** samples exactly whenever the weights are uniform. The discussion after Table 4 makes this precise.

2 Setup

Base model: kuleshov-group/udlm-qm9 (UDLM, uniform diffusion, `parameterization=d3pm`), loaded from HuggingFace with no local fine-tuning. Sequence length 32, 32 sampling steps, `use_cache=False`, seed 1. Rewards are computed with RDKit on decoded SMILES; candidates RDKit cannot parse receive $r = 0$, which is the minimum of both QED and ring count and therefore also penalizes invalidity. Every configuration below shares this base model, so differences are attributable to the guidance mechanism alone.

Metric conventions. The paper’s Table 5 reports raw *counts* out of 1024 generated samples (`Valid` = 994.8 means 994.8 of 1024 parsed), and its `Mean` column is the property averaged over the *novel* sequences only. The repository’s evaluation code instead stores fractions, and takes `Unique` and `Novel` as fractions *of the valid set* rather than of all samples. The two are consistent once converted: our unguided reference produces 1002/1024 valid and 125 novel with a novel QED mean of 0.465, against the paper’s UDLM validity of 1013.6 and data QED of 0.47. Tables 1 and 2 below are recorded in the repository’s percentage convention because that is what the runs printed; `scripts/make_results_table.py` converts to the paper’s counts, so the generated `sweep_table` files are the ones to compare against Table 5.

3 Which mixture-sampling variant to use

Table 1 compares the three variants at $N = 10$ on 512 samples. ESS is the effective sample size $1/\sum_n w_n^2$ of the importance weights averaged over the batch, reported at the final step; `ESS = N` means uniform weights and `ESS = 1` means the mixture collapsed onto a single candidate.

Table 1: Mixture-sampling variants, QED reward, $N = 10$, 512 samples.

Variant	λ	Valid %	Unique %	Novel %	QED mean	Novel QED mean	ESS
aggregate_x0	0	96.68	100.00	7.27	0.469	0.460	10.00
exact	50	85.55	99.09	25.80	0.511	0.512	4.29
marginal	50	84.18	99.77	22.97	0.501	0.484	4.90
aggregate_x0	50	95.31	99.80	9.02	0.487	0.451	4.63

Two findings drive the choice of default. First, **exact** attains the largest QED gain (0.469 $\rightarrow$ 0.511) but costs roughly 11 points of validity, whereas **aggregate_x0** keeps validity essentially at the unguided level (96.7% $\rightarrow$ 95.3%) for a smaller QED gain. Second, and more diagnostically, **exact** at $\lambda = 0$ does *not* recover the unguided sampler: it reaches only 61.7% validity (128-sample

run, Table 2) against 96.7% for **aggregate_x0**. Committing each step to a single sampled $\mathbf{x}_0$ injects the denoiser’s sampling noise into the trajectory, and at $N = 10$ that cost outweighs the extra fidelity to Eq. (3). Subsequent experiments therefore fix **aggregate_x0** and raise N to 30.

This conclusion is superseded. Both findings survive as observations at $N = 10$, but the inference drawn from them does not: at $N = 100$ –500 and $\lambda = 5000$, **exact** matches **marginal** on validity and gives the best novel QED and the highest novel count in this report. The validity cost identified here is real but is repaired by the tilt itself, for the reason given after Table 4. Read this section as the record of why **aggregate_x0** became the default, not as a live recommendation.

4 Exploratory λ sweeps

Table 2 records the first sweeps, run with **exact** and $N = 10$ on only 128 samples. They establish that the tilt does what the derivation says — QED rises monotonically in λ — but the sample size is too small to resolve the ring-count trend, where the standard error of the mean is roughly 0.1.

Table 2: λ sweeps, **exact** variant, $N = 10$, 128 samples each. Underpowered; superseded by the runs in Table 1 and the sweep described in Section 11.

Reward	λ	Valid %	Novel %	Property mean	Property median
QED	0	61.72	30.38	0.476	0.486
QED	10	87.50	22.32	0.493	0.488
QED	50	82.03	28.57	0.520	0.520
QED	200	81.25	34.62	0.530	0.538
ring count	0	61.72	30.38	1.608	1.000
ring count	2	77.34	33.33	1.788	2.000
ring count	10	85.16	36.70	1.651	1.000

The non-monotone ring-count column is consistent with weight degeneracy: ring count is integer-valued, so a unit difference in reward gives a weight ratio of e^λ , and at $\lambda = 10$ that is $\approx 2.2 \times 10^4$. The mixture then collapses onto whichever single candidate happens to score highest. This is what motivated logging ESS.

5 The (N, λ) grid

Table 3 is the main result so far: **aggregate_x0**, 128 samples per cell, guidance active over the whole trajectory. Counts follow the paper’s convention, and *novel QED* is the paper’s **Mean** column (property averaged over novel sequences only).

Three things stand out.

Both N and λ matter, and N is the harder constraint. Novel QED climbs from the unguided 0.465 to 0.532, and the gain requires *both* a large N and a large λ : at $N = 10$ even $\lambda = 1000$ only reaches 0.494, while at $N \geq 50$ the same λ reaches 0.528. This is the expected behaviour of self-normalized importance sampling — it can only reweight among the N proposals drawn from p_θ , never invent a better $\mathbf{x}_0$, so N sets the ceiling and λ only controls how sharply we move toward it. Raising λ at small N buys nothing because the best of 10 draws is not much better than the mean.

Table 3: (N, λ) grid, `aggregate_x0`, 128 samples per cell. Novel QED is the metric comparable to Table 5 of the paper. The reference rows are the QM9 training set and our unguided sampler at 1024 samples.

N	λ	Valid	Unique	Novel	QED mean	Novel QED mean
QM9 data (133,885)		133,885	133,885	133,885	0.465	0.465
Unguided (1024)		1002	999	125	0.471	0.465
10	50	120	119	9	0.479	0.462
10	200	121	120	17	0.498	0.476
10	1000	121	121	26	0.505	0.494
30	50	126	126	14	0.482	0.438
30	200	124	124	17	0.506	0.479
30	1000	122	122	31	0.511	0.507
50	50	126	126	12	0.505	0.510
50	200	124	123	15	0.524	0.523
50	1000	121	121	24	0.532	0.528
100	50	126	126	14	0.493	0.485
100	200	126	126	23	0.522	0.532
100	1000	123	123	32	0.526	0.524

Validity is preserved. Every cell keeps 94–98% validity, against 97.9% unguided. This is the payoff for fixing `aggregate_x0`: the `exact` variant bought a comparable QED gain but gave up 11 points of validity (Table 1).

We are still short of the published baseline. Table 5 of the paper reports novel QED = 0.61 for UDLM + D-CBG. Our best is 0.532. D-CBG perturbs the per-token logits with a classifier gradient, which can push probability toward regions the proposal distribution essentially never visits; importance sampling cannot. Closing the gap therefore needs either much larger N , or a proposal that is itself tilted rather than reweighted after the fact.

Caveat on precision. At 128 samples only 9–32 molecules per cell are novel, so with a QED spread of roughly 0.1 the standard error on novel QED is about 0.02–0.03. Individual cells differing by 0.02 are not separated; the monotone trends across the grid are, since they are consistent across many cells. The $N = 30$, $\lambda = 50$ cell reading 0.438 (below unguided) is the clearest instance of this noise. The configurations in Section 11 rerun the promising settings at 1024 samples.

6 Head-to-head against D-CBG

Table 4 is the comparison that matters: every row uses the same HuggingFace UDLM-QM9 base model, 1024 samples, seed 1, 32 steps. Novel QED is the column comparable to Table 5 of the paper.

The marginal variant beats `aggregate_x0` at every matched setting. The last block reruns the four *Ours* configurations with `mixture_sampling=marginal`, which averages the N nor-

Table 4: Same base model, same budget, 1024 samples. *Novel QED* is the paper’s **Mean** column (property over novel sequences only). Counts are out of 1024. The *Ours* block uses `aggregate_x0`; the *Ours (marginal)* and *Ours (exact)* blocks repeat the same four settings with `mixture_sampling=marginal` and `=exact`. All three blocks share the identical budget, so the blocks differ only in how the N weighted candidates are turned into a reverse step.

Method	Setting	Valid	Unique	Novel	QED mean	Novel QED
QM9 data	—	—	—	—	0.465	0.465
Unguided UDL	—	1002	999	125	0.471	0.465
D-CBG (approx.)	$\gamma = 1$	997	993	121	0.481	0.477
D-CBG (approx.)	$\gamma = 2$	976	972	116	0.499	0.485
D-CBG (approx.)	$\gamma = 5$	856	830	78	0.556	0.529
D-CBG (exact)	$\gamma = 1$	949	937	118	0.542	0.528
D-CBG (exact)	$\gamma = 2$	903	879	115	0.581	0.567
D-CBG (exact)	$\gamma = 5$	968	842	82	0.600	0.593
Ours	$N=100, \lambda=5000$	970	946	262	0.529	0.522
Ours	$N=300, \lambda=5000$	947	909	289	0.537	0.530
Ours	$N=500, \lambda=5000$	949	911	321	0.542	0.534
Ours	$N=500, \lambda=1000, t \leq 0.75$	946	925	251	0.540	0.535
Ours (marginal)	$N=100, \lambda=5000$	932	905	287	0.536	0.527
Ours (marginal)	$N=300, \lambda=5000$	923	888	307	0.541	0.534
Ours (marginal)	$N=500, \lambda=5000$	915	882	334	0.548	0.541
Ours (marginal)	$N=500, \lambda=1000, t \leq 0.75$	928	904	303	0.545	0.544
Ours (exact)	$N=100, \lambda=5000$	920	891	300	0.540	0.534
Ours (exact)	$N=300, \lambda=5000$	936	902	325	0.544	0.541
Ours (exact)	$N=500, \lambda=5000$	921	885	369	0.548	0.549
Ours (exact)	$N=500, \lambda=1000, t \leq 0.75$	930	896	342	0.547	0.543

marginalized posteriors $q(\cdot \mid x_t, x_0^{(n)})$ instead of substituting the weighted candidate histogram $\bar{x}_0$ into the posterior expression. Under uniform diffusion the two are not equivalent — the posterior normalizer depends on x_0 , so `aggregate_x0` normalizes once where the mixture normalizes per component — and the difference is systematic: novel QED improves in every row (by +0.004 to +0.009), novel counts rise to 287–334 against 251–321, and the price is 2–4 points of validity (89–91 % against 92–95 %). The best marginal cell, $N=500, \lambda=1000, t \leq 0.75$ at 0.544, becomes our strongest novel-QED result, though it still trails exact D-CBG’s 0.593. Wall-clock cost is unchanged: the sampler is RDKit-bound, and the extra N -fold posterior average is negligible on GPU.

Neither block in Table 4 samples Eq. (3). Both are approximations of it, and the approximation runs the opposite way to how the variants were introduced, so it is worth stating precisely. Equation (3) is a mixture of N posteriors $q(\cdot \mid \mathbf{x}_t, \mathbf{x}_0^{(n)})$, each of which factorizes over positions; a mixture of product measures is not a product measure. **exact** draws $n^* \sim \text{Cat}(w)$ and then samples that one component, which is ancestral sampling of the mixture — it is the only one of the three that targets Eq. (3) itself, with no error beyond the finite- N bias of the self-normalized weights and p_θ ’s own error. **marginal** forms $\sum_n w_n q(\cdot \mid \mathbf{x}_t, \mathbf{x}_0^{(n)})$ and samples positions independently, so

it draws from $\prod_{\ell} (\sum_n w_n q_{n,\ell})$ where the target is $\sum_n w_n \prod_{\ell} q_{n,\ell}$: the same per-position marginals, a different joint. Note also what **exact** does *not* throw away. The $N - 1$ unselected candidates set the softmax normalizer and therefore the selection probabilities, so all N rewards are used; what **exact** declines to do is average the posteriors, and that average is exactly the operation that destroys the cross-position correlations.

The gap is not cosmetic. Take absorbing-state diffusion, $L = 2$ with both positions masked, $N = 2$ with uniform weights, candidates $\mathbf{x}_0^{(1)} = (a, a)$ and $\mathbf{x}_0^{(2)} = (b, b)$, and probability $\frac{1}{2}$ of unmasking each position. The mixture gives (a, b) probability zero, since either component commits both positions to the same letter. **marginal**, whose per-position marginal is $\{a : \frac{1}{4}, b : \frac{1}{4}, \text{mask} : \frac{1}{2}\}$, gives it $\frac{1}{16}$. Sampling positions independently lets the sampler splice fragments of different candidates into sequences no component proposes. **aggregate_x0** carries a second error on top of this under uniform diffusion: because the posterior normalizer depends on $\mathbf{x}_0$, substituting $\bar{\mathbf{x}}_0$ and normalizing once does not even reproduce the mixture’s per-position marginals.

Fidelity to Eq. (3) tracks novel QED, but not validity. Ordering the variants by how little they approximate — **exact**, then **marginal**, then **aggregate_x0** — was recorded here as a two-part prediction before the **exact** rows were run: best novel QED and worst validity. The first half held and the second did not.

Novel QED follows the fidelity order in three of the four matched settings, with **exact** above **marginal** by +0.007, +0.007 and +0.008 at $N = 100, 300$ and 500 ($\lambda = 5000$), and level with it in the fourth (0.543 against 0.544 at $N=500, \lambda=1000, t \leq 0.75$). As with the **marginal**-versus-**aggregate_x0** comparison, the per-cell standard error is about 0.005, so no single row separates the two; the claim rests on three rows moving the same way by more than the step between the other two variants. Novel counts are cleaner: **exact** produces 300–369 novel molecules against 287–334 for **marginal** and 251–321 for **aggregate_x0**, winning every matched row, and its 369 at $N=500, \lambda=5000$ is the highest novel count anywhere in this report. Its 0.549 there is our best novel QED, displacing **marginal**’s 0.544.

Validity refutes the prediction. **exact** lands at 920, 936, 921 and 930 against **marginal**’s 932, 923, 915 and 928 — differences of $-12, +13, +6, +2$, with no consistent sign. The two are indistinguishable, and both sit two to three points below **aggregate_x0**. Committing each step to a single sampled $\mathbf{x}_0$ therefore costs nothing in validity once the tilt is on, which is not what the $\lambda = 0$ control or Table 1 would suggest; the next paragraph explains why. The rejection of **exact** in Section 3 on $N = 10$ evidence does not survive at $N \geq 100$.

What the $\lambda = 0$ control measures. At $\lambda = 0$ the weights are exactly uniform, so **exact** reduces to drawing a single $\mathbf{x}_0 \sim p_{\theta}(\mathbf{x}_0 | \mathbf{x}_t)$ and sampling its posterior: ancestral sampling of the *true* untilted reverse step, with no Monte Carlo error and no dependence on N . The base sampler instead plugs the denoiser marginal $\mathbf{x}_{\theta}$ into the posterior and samples positions independently, which is the factorized projection of that same step. The control reproduces at 1024 samples: 679 valid (66.3%) against 1002 (97.9%) for the base sampler, with QED 0.468 and novel QED 0.457, i.e. the unguided property level. The measured ESS is exactly 100 at $N = 100$, confirming the weights are uniform and that the row isolates the sampler rather than the tilt. Table 2’s 61.7% at 128 samples was therefore not small-sample noise.

This is not a defect of **exact**. It says that on this checkpoint the factorized projection *helps*, and that sampling the true reverse step costs 31.6 points of validity before any guidance is applied — the projection acts as a regularizer that the released UDLM sampler is implicitly tuned around. It also corrects the reading in Section 1, which presented the failure to recover unguided sampling

as something **exact** lacks: it is the base sampler that does not sample its own reverse step.

The tilt repairs the validity the exact sampler costs. Putting the control next to the guided rows resolves the contradiction in the previous paragraph. **exact** at $\lambda = 0$ is 66.3 % valid; the same $N = 100$ with $\lambda = 5000$ is 89.8 %. The tilt does not merely fail to cost validity, it buys back 23.5 of the 31.6 points the sampler gave up. The mechanism is the invalid-candidate convention of Section 1: RDKit-unparseable candidates receive $r = 0$, the minimum of the reward range, so $\text{softmax}(\lambda r)$ actively selects against exactly the candidates whose commitment produces invalid molecules. At $\lambda = 5000$ that selection is strong enough to undo most of the damage. Two consequences follow. The reward is doing double duty — property optimization and validity repair — so a reward that did *not* bottom out on invalid inputs would leave **exact** at its $\lambda = 0$ validity, and **exact** would then be the wrong variant. And the $N = 10$ evidence in Section 3 was misread at the time: it attributed the validity loss to committing to a single $\mathbf{x}_0$, which is right, but concluded the loss was unrecoverable, which required a λ large enough to matter and $N = 10$ could not supply one.

Relation to EDLM. The **exact** variant is Algorithm 1 of EDLM (*Denoising via Importance Sampling*) with the learned energy $E_\phi(\mathbf{x}_0, \mathbf{x}_t)$ replaced by $-\lambda r(\mathbf{x}_0)$: draw k candidates from the denoiser, score them, resample one with probability proportional to $\exp(-e^i)$, and step through the exact posterior of the survivor. Two differences matter. First, EDLM’s energy conditions on $\mathbf{x}_t$ and has to be trained, whereas r depends on $\mathbf{x}_0$ alone, which is what makes the importance ratio available in closed form (Eq. 2). Second, EDLM’s residual energy is small by construction, so its target sits close to the proposal and $k = 8$ suffices; a reward tilt at useful λ places the target further away, and N has to grow with λ to compensate.

How far is roughly measurable. Over the $\lambda = 5000$ runs the median ESS is about 31 at $N = 100$, 52 at $N = 300$ and 62 at $N = 500$, with per-step ranges of 8.6–67.5, 10.3–113 and 1.0–141. An earlier draft of this section asserted that ESS collapses toward 1 at $\lambda = 5000$, making the sampler equivalent to best-of- N selection. That is wrong: steps with $\text{ESS} < 5$ are 1.1 % of the total at $N = 500$ and absent at $N \leq 300$, and a median in the tens means many candidates carry real weight. What is true is weaker and still useful — ESS grows sublinearly in N , from about 31 % of N at $N = 100$ to 12 % at $N = 500$, so each additional candidate buys less than the one before, which is the honest form of the diminishing returns visible across the N rows of Table 4.

Two caveats on these numbers. They are recovered from the sampler’s progress line, which does not render every step: the fastest configuration logged only 272 of its 528 step-cells, so its statistics are the least trustworthy of the four and the medians should be read to two significant figures at most. And ESS counts duplicate candidates as distinct samples. Once a sequence is largely determined the N proposals repeat heavily — which is why the reward cache in `our_guidance.py` pays off at all — and identical candidates carry identical rewards, so part of the ESS reported here is tied weights on the same molecule rather than genuine spread. ESS is therefore an upper bound on the diversity actually being reweighted, and λ has to be chosen by outcome rather than by ESS. Table 8 does exactly that. EDLM’s experiments are on masked diffusion; the runs here are uniform, so the **exact** rows also test whether the substitution transfers, and on this evidence it does. Credit for the resampling scheme belongs to EDLM, and **marginal** and **aggregate_x0** are the variants specific to this work.

D-CBG wins on the property; we win on novel throughput. The exact D-CBG variant is far stronger than its first-order approximation (0.593 against 0.529 at $\gamma = 5$), and it beats our

best configuration on novel QED by a clear margin, 0.593 against 0.549 (**exact**, $N=500$, $\lambda=5000$). Any earlier reading of these experiments based on the approximate branch alone — which briefly suggested our method was ahead on every axis — was an artifact of comparing against the weaker variant, and does not survive.

What our method retains is throughput of *new* molecules at matched validity. D-CBG reaches 0.593 with 82 novel molecules out of 1024, fewer than unguided sampling produces (125), and its uniqueness drops to 87% — it is concentrating hard onto a small high-QED region. Ours produces 251–369 novel molecules, two to four and a half times as many, at 90–95% validity and 96–98% uniqueness. The **exact** variant is the strongest on this axis as well as on novel QED: 369 novel molecules at 90% validity, against D-CBG’s 82 at 95%. Which matters depends on the use case: for “enumerate many new candidates above baseline quality” ours is clearly better, and Section 6.1 shows that even “find the single most drug-like molecule” does not go to D-CBG once the comparison is made at matched sampling budget rather than on the mean.

The other standing advantage is operational rather than metric: our method requires no trained component. D-CBG needed a 25k-step noisy classifier (3.5 hours here), and its exact variant needs $L \cdot V$ classifier forwards per denoising step, which exhausted a 3090’s memory at batch size 64 and had to be re-run at batch size 8. Ours needs only RDKit.

Caveat on the published number. The paper reports 0.61 for UDLM + D-CBG; our exact reproduction reaches 0.593, close enough that the remaining gap is plausibly the base checkpoint (they train their own UDLM rather than using the HuggingFace release), 5-seed averaging, and classifier tuning. The approximate branch at 0.529 is evidently not what the paper reports.

6.1 Past the mean: the QED distribution behind Table 4

Both property columns of Table 4 are means, and a mean is a poor summary when the methods being compared differ in how much spread they leave. `our_qm9_eval.py` dumps the per-sample QED of every valid and novel molecule, so the same 1024-sample runs can be re-read as distributions without any new sampling. Table 5 does that over the novel set: quartiles, the maximum, the mean of the top 20 novel molecules — matched on the generation budget, 20 out of the same 1024 samples for every row — and the number of novel molecules clearing $\text{QED} = 0.6$, a level well above the QM9 upper quartile of 0.515 that unguided sampling clears twice among the 125 novel molecules it produces per 1024 draws.

D-CBG’s mean advantage is a lower-tail effect. At $\gamma = 5$ exact D-CBG leads our best row by 0.044 on the mean (0.593 against 0.549), but the lead is not uniform over the distribution. It is 0.065 at the first quartile (0.575 against 0.510), 0.050 at the median, 0.026 at the third quartile, and it reverses to -0.019 at the top 20. Table 6 states this directly by averaging within each quartile bin: D-CBG at $\gamma = 5$ spans only 0.101 from its bottom quarter to its top quarter, against 0.155 for our **exact** $N=500$ row and 0.162 for unguided sampling. Raising γ compresses the distribution into a narrow band just below 0.6; it does not extend its upper end. Our tilt shifts the whole distribution up and keeps roughly the unguided spread, which costs the mean — our bottom quarter averages 0.471 against D-CBG’s 0.537 — and costs nothing at the top.

At the top of the distribution the ordering reverses. On the 20 best novel molecules out of the same 1024-sample budget we reach 0.657 (**marginal** $N=300$ and **exact** $N=500$, which are indistinguishable from each other) against 0.638 for D-CBG exact $\gamma = 5$: $+0.019$ with a standard error of 0.005 on the difference, about 3.7σ , so unlike the 0.005-level differences between our own

Table 5: QED distribution over the novel set, same runs as Table 4. *Novel* and > 0.6 are counts out of 1024 generated samples; *Top-20* is the mean of the 20 best novel molecules, so it is matched on generation budget rather than on novel count. *Max*₇₈ is the mean maximum of a random 78-molecule subsample (4000 draws, 78 = the smallest novel count in the table), which is the max column made comparable across rows — the raw *Max* is not, since it grows with the novel count and the rows differ by $4.7\times$ in it. Bold marks the best value in each column at printed precision.

Method	Setting	Novel	Mean	Q1	Med.	Q3	Max	Max ₇₈	Top-20	> 0.6
QM9 data	—	—	0.465	0.427	0.473	0.515	—	—	—	—
Unguided UDL	—	125	0.465	0.430	0.468	0.504	0.628	0.618	0.558	2
D-CBG (approx.)	$\gamma = 1$	121	0.477	0.437	0.474	0.532	0.643	0.635	0.582	4
D-CBG (approx.)	$\gamma = 2$	116	0.485	0.447	0.496	0.537	0.619	0.618	0.588	6
D-CBG (approx.)	$\gamma = 5$	78	0.529	0.479	0.541	0.585	0.658	0.658	0.610	12
D-CBG (exact)	$\gamma = 1$	118	0.528	0.484	0.544	0.581	0.689	0.683	0.628	19
D-CBG (exact)	$\gamma = 2$	115	0.567	0.541	0.568	0.600	0.654	0.654	0.633	29
D-CBG (exact)	$\gamma = 5$	82	0.593	0.575	0.600	0.622	0.656	0.656	0.638	42
Ours	$N=100, \lambda=5000$	262	0.522	0.489	0.525	0.566	0.665	0.655	0.635	35
Ours	$N=300, \lambda=5000$	289	0.530	0.491	0.530	0.572	0.660	0.653	0.635	32
Ours	$N=500, \lambda=5000$	321	0.534	0.497	0.533	0.584	0.687	0.665	0.646	50
Ours	$N=500, \lambda=1000, t \leq 0.75$	251	0.535	0.498	0.534	0.575	0.697	0.685	0.649	33
Ours (marginal)	$N=100, \lambda=5000$	287	0.527	0.491	0.528	0.568	0.697	0.675	0.642	30
Ours (marginal)	$N=300, \lambda=5000$	307	0.534	0.500	0.529	0.576	0.717	0.687	0.657	50
Ours (marginal)	$N=500, \lambda=5000$	334	0.541	0.500	0.545	0.586	0.687	0.668	0.652	61
Ours (marginal)	$N=500, \lambda=1000, t \leq 0.75$	303	0.544	0.500	0.548	0.595	0.675	0.667	0.650	65
Ours (exact)	$N=100, \lambda=5000$	300	0.534	0.496	0.539	0.580	0.672	0.654	0.638	42
Ours (exact)	$N=300, \lambda=5000$	325	0.541	0.507	0.542	0.590	0.677	0.661	0.647	58
Ours (exact)	$N=500, \lambda=5000$	369	0.549	0.510	0.550	0.596	0.735	0.678	0.657	81
Ours (exact)	$N=500, \lambda=1000, t \leq 0.75$	342	0.543	0.505	0.545	0.583	0.712	0.681	0.656	65

variants this one is resolved by a single seed. The best molecule found is 0.735 for us against 0.689 for the best D-CBG row ($\gamma = 1$) and 0.656 at $\gamma = 5$. And the count that matters if the sampler is a candidate generator: 81 novel molecules above 0.6 per 1024 samples against 42, $7.9 \pm 0.8\%$ against $4.1 \pm 0.6\%$, nearly double at the same cost. That last comparison is the only one where our weaker cells do not hold up: the **exact** block matches or beats D-CBG’s best row in every one of its four settings (42, 58, 81, 65), but across all twelve of our configurations the count ranges from 30 to 81, so the low- N **aggregate_x0** and **marginal** cells fall below the 42 of $\gamma = 5$.

The raw *Max* column overstates this, and the *Max*₇₈ column is the honest version: the maximum of a sample grows with the sample size, and we have $3\text{--}4.5\times$ as many novel molecules to take a maximum over. Matched at 78 novel molecules our best rows fall to 0.678–0.687 while D-CBG exact $\gamma = 1$ holds 0.683, so at equal *novel count* the best single molecule is a wash between the two methods, and both beat $\gamma = 5$ ’s 0.656. What survives matching is the budget-matched statement rather than the count-matched one: for a fixed number of *sampler calls* — which is what a practitioner pays for — our upper tail is strictly better, because we convert that budget into more novel molecules in the first place. The one-line summary is that increasing γ trades the upper tail for the lower tail, whereas increasing N moves both.

Figure 1 plots the two readings side by side. The mean view (a) is the one Table 4 reports and it shows a real trade-off: the front runs from D-CBG’s high-quality, low-novelty corner to our high-novelty corner, and picking a point on it is a choice about what the samples are for. The

Table 6: Mean novel QED within each quartile bin, and the gap between the top and bottom bins (*Spread*). Guidance strength buys D-CBG its mean by shrinking the spread; ours preserves it.

Method	Setting	Bottom 25%	25–50%	50–75%	Top 25%	Spread
Unguided UDLM	—	0.380	0.452	0.488	0.542	0.162
D-CBG (approx.)	$\gamma = 1$	0.382	0.456	0.506	0.568	0.186
D-CBG (approx.)	$\gamma = 2$	0.376	0.473	0.515	0.575	0.200
D-CBG (approx.)	$\gamma = 5$	0.433	0.515	0.563	0.611	0.179
D-CBG (exact)	$\gamma = 1$	0.418	0.522	0.562	0.617	0.199
D-CBG (exact)	$\gamma = 2$	0.498	0.560	0.586	0.625	0.127
D-CBG (exact)	$\gamma = 5$	0.537	0.589	0.612	0.638	0.100
Ours	$N=100, \lambda=5000$	0.433	0.508	0.544	0.604	0.172
Ours	$N=300, \lambda=5000$	0.454	0.511	0.552	0.603	0.149
Ours	$N=500, \lambda=5000$	0.453	0.513	0.558	0.614	0.161
Ours	$N=500, \lambda=1000, t \leq 0.75$	0.458	0.514	0.556	0.614	0.156
Ours (marginal)	$N=100, \lambda=5000$	0.446	0.513	0.550	0.602	0.156
Ours (marginal)	$N=300, \lambda=5000$	0.449	0.514	0.556	0.618	0.168
Ours (marginal)	$N=500, \lambda=5000$	0.459	0.523	0.566	0.619	0.160
Ours (marginal)	$N=500, \lambda=1000, t \leq 0.75$	0.460	0.524	0.571	0.623	0.163
Ours (exact)	$N=100, \lambda=5000$	0.445	0.521	0.559	0.610	0.165
Ours (exact)	$N=300, \lambda=5000$	0.457	0.522	0.566	0.618	0.161
Ours (exact)	$N=500, \lambda=5000$	0.471	0.527	0.572	0.626	0.155
Ours (exact)	$N=500, \lambda=1000, t \leq 0.75$	0.463	0.526	0.563	0.620	0.157

top-20 view (b) is not a trade-off at all — our configurations dominate the D-CBG family outright, since they are better on both axes simultaneously. The two panels use identical data and differ only in which order statistic of the same QED distribution is on the y axis, which is the point: the head-to-head verdict of Section 6 is a statement about means, not about quality.

Figure 2 shows the distributions the two summaries come from, and the two families move differently. For D-CBG the box climbs toward a ceiling that does not move with it: from $\gamma = 1$ to $\gamma = 5$ the median goes $0.544 \rightarrow 0.600$ while the 95th percentile only goes $0.637 \rightarrow 0.648$ and the 99th actually falls, $0.677 \rightarrow 0.654$, as does the maximum, $0.689 \rightarrow 0.656$. Guidance strength is buying the middle of the distribution by clipping its extreme tail. For ours the tail is what moves: from $N = 100$ to $N = 500$ the 95th percentile goes $0.628 \rightarrow 0.643$, the 99th $0.652 \rightarrow 0.657$, and the maximum $0.672 \rightarrow 0.735$, with the median rising only $0.539 \rightarrow 0.550$.

What this does and does not change. It does not overturn Section 6: on the paper’s metric — mean QED over novel sequences — exact D-CBG is ahead by 0.044 and that number stands. What it changes is the interpretation. The gap is not evidence that D-CBG’s guidance finds better molecules; it is evidence that D-CBG’s guidance is more *aggressive*, moving mass off the low-QED tail at the cost of novelty and diversity, and a mean rewards exactly that. On any statistic that asks about the good molecules rather than the average one — top-20, count above a threshold, best-of-budget — the reward-tilted posterior is ahead at the same sampling cost and with no trained classifier. Both readings are worth reporting, and a single-number comparison on QED mean is the one that flatters the baseline.

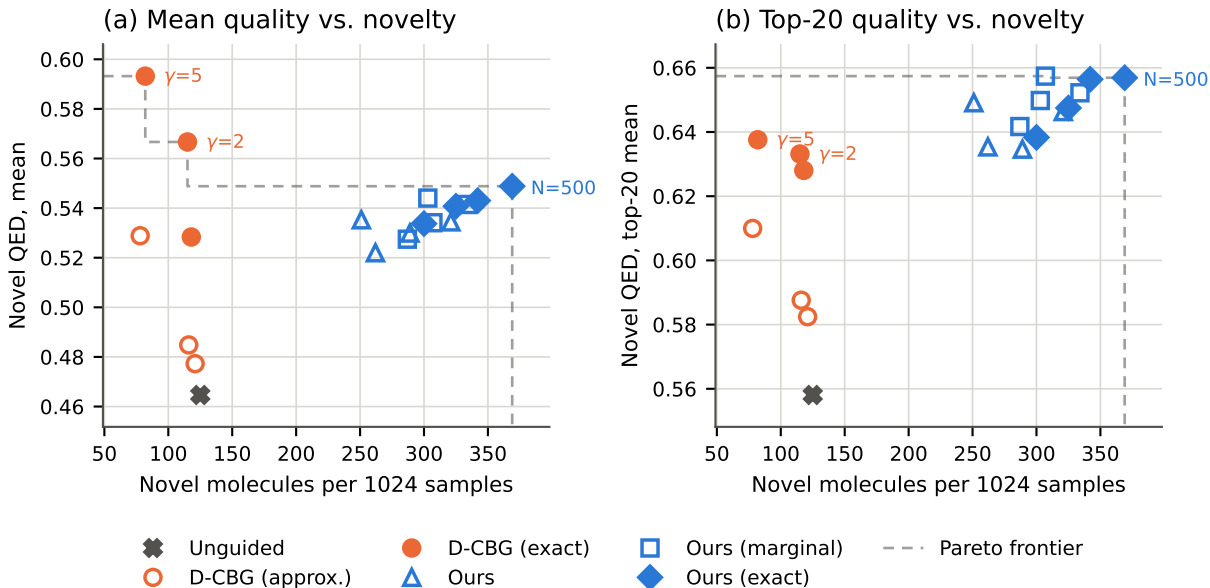


Figure 1: Quality against novelty, all 19 configurations of Table 4. Panel (a) uses the mean novel QED of that table: the front holds three points — D-CBG exact $\gamma = 5$ and $\gamma = 2$, and our **exact** $N=500$ — so on the mean neither method dominates and the choice is a genuine trade-off; note that D-CBG exact $\gamma = 1$ and the whole approximate branch are dominated by our rows. Panel (b) replaces the mean with the budget-matched top-20 mean and the picture changes: the front is entirely ours, every D-CBG configuration is strictly dominated, and 10 of our 12 configurations beat $\gamma = 5$ on *both* axes at once (the remaining two match its top-20 to within 0.003 with 3.2–3.5 $\times$ the novel count).

7 Denoising steps: 32 against 150

Every run above fixes `sampling.steps` = 32, the paper’s setting. The step count deserves its own axis because the reward-tilted posterior draws a fresh set of N candidates and applies the tilt at *every* step, so 150 steps apply it 4.7 times as often at the same N ; and because **exact** commits to one sampled $\mathbf{x}_0$ per step, so more steps means more commitments but a smaller move per commitment. Table 7 repeats the two variants at 150 steps, extending N down to 10 and 50, with an unguided row at the same step count as reference. $T = 0$ (continuous time), so there is no t -grid snapping and 150 steps is well posed. `aggregate_x0` and D-CBG are omitted; D-CBG’s exact branch costs about 55 minutes per configuration at 32 steps and would be roughly 4.3 hours at 150.

More steps buy validity we did not need and give up the throughput advantage. The direction is the same in all eight matched pairs: validity rises by 49–82 counts, novel counts fall by 59–117, and novel QED moves by at most ± 0.010 with seven of eight pairs flat or slightly negative. Guided validity goes from 89–91% at 32 steps to 96–99% at 150, which closes the gap to `aggregate_x0` and to unguided sampling entirely — but validity was never the binding constraint, since novel QED is the reported metric and D-CBG already matched us there. What is lost is the one axis where we beat D-CBG outright: our best novel count falls from 369 to 280, against D-CBG’s 82. At 4.7 times the sampling cost, 150 steps is a worse operating point for this method,

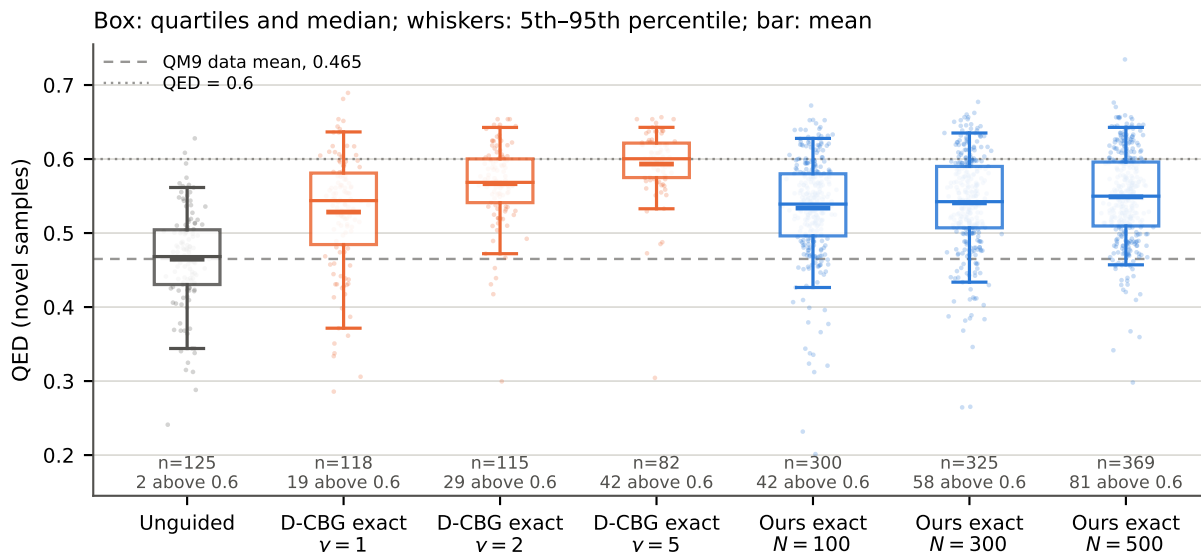


Figure 2: Novel QED distributions behind the two means, exact variants only, one dot per novel molecule. Increasing γ (orange) concentrates the distribution into a band below 0.6 and shrinks the novel count; increasing N (blue) shifts the distribution up while keeping its spread, so the count above 0.6 grows from 42 to 81 even though the mean stays lower.

and 32 steps should stay the default.

The effect is specific to the guided sampler. Unguided sampling moves the *opposite* way: 125 novel molecules at 32 steps against 165 at 150, an increase, while every guided row decreases. So this is not a generic “more steps means more memorization” effect of the base model. The reading that fits both is that step size, not step count, controls how far guidance displaces a trajectory. At 32 steps each reverse step is a large move and the tilt pushes samples well off the model’s typical set — high novelty, low validity, and the highest novel QED. At 150 steps each move is small, the tilt displaces less even though it is applied 4.7 times as often, and samples stay near the typical set — high validity, low novelty. The net displacement shrinks as steps grow, which is the same quantity the λ sweep below saturates against.

The exact-versus-marginal ordering washes out. At 32 steps **exact** beat **marginal** on novel QED in three of four matched rows by a consistent +0.007 to +0.008. At 150 steps it wins four of six but with mixed signs and a mean advantage of about +0.002: −0.007 at $N=10$, +0.009 at $N=50$, −0.003 at $N=100$, +0.005 at $N=300$, +0.006 at $N=500$ and +0.002 on the window row. Against a per-cell standard error near 0.005 that is a wash. This is consistent with the step-size reading: the two variants differ only in whether cross-position correlations survive within a single reverse step, and when each step moves less there is less correlation to preserve or destroy. The distinction that Section 1 establishes as mathematically real is therefore also empirically small whenever the sampler takes small steps — which is worth knowing before treating **exact** as the variant to standardize on.

The λ curve saturates far earlier at 150 steps. $\lambda = 5000$ was selected on the 32-step `aggregate_x0` grid (Section 5) and never retuned for either variant or for a different step count.

Table 7: 150 denoising steps, 1024 samples, seed 1, same UDLM-QM9 base model. Δ columns are the change from the matched 32-step row of Table 4; a dash means Table 4 has no such row. All settings use $\lambda = 5000$ over all t except the last row of each block.

Variant	Setting	Valid	Novel	Novel QED	Δ Valid	Δ Novel	Δ NQED
Unguided	—	1015	165	0.459	+13	+40	−0.006
exact	$N=10$	995	199	0.500	—	—	—
exact	$N=50$	1011	193	0.530	—	—	—
exact	$N=100$	1002	208	0.531	+82	−92	−0.003
exact	$N=300$	985	250	0.538	+49	−75	−0.003
exact	$N=500$	994	252	0.543	+73	−117	−0.006
exact	$N=500, \lambda=1000, t \leq 0.75$	991	280	0.536	+61	−62	−0.007
marginal	$N=10$	992	219	0.507	—	—	—
marginal	$N=50$	1001	210	0.521	—	—	—
marginal	$N=100$	1005	222	0.534	+73	−65	+0.007
marginal	$N=300$	996	248	0.533	+73	−59	−0.001
marginal	$N=500$	984	254	0.537	+69	−80	−0.004
marginal	$N=500, \lambda=1000, t \leq 0.75$	980	236	0.534	+52	−67	−0.010

Table 8 probes it at 150 steps.

Table 8: λ at 150 steps, **exact**, $N = 100$, 1024 samples. The 32-step column is the `aggregate_x0` grid at the same N (Table 3), which is the only λ series available at 32 steps.

λ	Valid	Unique	Novel	Novel QED	Novel QED (32 steps)
100	1011	995	211	0.524	—
200	—	—	—	—	0.505
500	998	963	193	0.531	—
1000	998	956	190	0.531	0.515
5000	1002	958	208	0.531	0.522

At 150 steps novel QED is 0.531 at $\lambda = 500, 1000$ and 5000 — flat to three decimals — and only $\lambda = 100$ falls off, to 0.524. At 32 steps the same quantity was still climbing at the top of the range ($0.505 \rightarrow 0.515 \rightarrow 0.522$ for $\lambda = 200 \rightarrow 1000 \rightarrow 5000$). So λ can be cut tenfold at 150 steps for nothing, but cutting it does not *improve* anything: the flat region is a plateau, not a peak, and the 32-step choice was not mistuned. Combined with the previous paragraphs, more steps flatten every knob this method exposes — λ , the variant, and N (novel QED spans 0.531–0.543 across $N = 100$ –500 at 150 steps against 0.534–0.549 at 32). The method has more headroom at coarse discretization, not less.

What this axis does not settle. There is no $\lambda = 0$ control at 150 steps. At 32 steps that row showed **exact** sampling the true reverse step at 66.3% validity against 97.9% for the factorized base sampler, and the tilt recovering 23.5 of the 31.6 lost points. Whether the 31.6-point gap itself shrinks with step size — it should, if step size is what governs displacement — is untested, and one 5-minute run would answer it.

8 Why the novel mean is the metric that matters

QED (Bickerton et al., 2012) is a weighted geometric mean of eight desirability functions over molecular weight, ALOGP, hydrogen-bond donors and acceptors, polar surface area, rotatable bonds, aromatic rings, and structural alerts. Its molecular-weight term peaks near 300 Da, whereas QM9 molecules have at most nine heavy atoms (≈ 100 –150 Da), so QM9 sits at 0.465 largely because its molecules are too small to score well. Raising QED on this dataset means generating larger, more complex molecules, which is also why validity falls as QED rises.

The valid set decomposes into molecules already present in QM9 and genuinely novel ones, and the two have systematically different QED. Splitting the generated samples accordingly:

Table 9: QED decomposed over the valid set. *Memorized* are valid samples already in QM9; *novel* are the rest. 1024 samples each. The gap between the two columns is what separates recall from generalization.

Run	Valid	Novel	QED (all)	QED memorized	QED novel	Gap
Unguided	1002	125	0.471	0.471	0.465	0.007
D-CBG approx., $\gamma = 5$	856	78	0.556	0.558	0.529	0.030
D-CBG exact, $\gamma = 1$	949	118	0.542	0.544	0.528	0.016
D-CBG exact, $\gamma = 2$	903	115	0.581	0.583	0.567	0.016
D-CBG exact, $\gamma = 5$	968	82	0.600	0.600	0.593	0.007
Ours $N=500$, $\lambda=1000$, $t \leq 0.75$	946	251	0.540	0.541	0.535	0.006
Ours $N=500$, $\lambda=5000$	949	321	0.542	0.545	0.534	0.010

Memorized molecules score higher than novel ones in every run, which is why the all-valid mean always exceeds the novel mean: the valid set is 75–90 % memorized, so the overall average is pulled toward it. Memorized molecules are real QM9 entries — curated, well-formed chemistry — while the novel ones are the model’s extrapolation, valid but less clean. (The duplicate-versus-set asymmetry in how the repository computes the two columns is *not* the cause: recomputing the novel mean with duplicates retained changes it by at most 0.002, since uniqueness runs 92–99 %.)

The gap is small and similar for most methods, with one exception: the *approximate* D-CBG branch at $\gamma = 5$ has a gap of 0.030, four to five times everyone else’s. Its memorized samples average 0.558 while its novel ones fall to 0.529, meaning much of what the first-order approximation buys is recovery of high-QED molecules already in the training set rather than new chemistry. The exact branch does not share this pathology — its gap at $\gamma = 5$ is 0.007, the same as unguided and as ours — so this is a property of the Taylor approximation, not of classifier guidance as such. An earlier version of this note generalized the approximate branch’s behaviour to D-CBG in general; that was wrong.

What survives is the count. At $\gamma = 5$ both D-CBG branches produce *fewer* novel molecules than unguided sampling (82 and 78 against 125), while our method produces 251–321. So the methods differ less in the quality of the novel molecules they make than in how many they make at all.

The decomposition also explains why the paper reports the novel mean rather than the overall mean: the overall mean is contaminated by how well a method retrieves memorized high-property molecules, while the novel mean measures whether guidance actually creates new ones.

9 The guidance time window

Table 10 restricts guidance to a sub-range of t , which counts down from 1 (pure noise) to 0 (clean). Outside the window the sampler falls back exactly to the base model and skips the reward calls entirely.

Table 10: Guidance time window, $\lambda = 1000$, 1024 samples. $t \leq 0.75$ means guidance is off for the 8 noisiest steps of 32; $t \geq 0.5$ means it is active *only* for those early steps.

Window	$N = 300$					$N = 500$				
	Valid	Novel	QED	Novel	QED	Valid	Novel	QED	Novel	QED
all t	944	229	0.535		0.526	967	276	0.538		0.530
$t \leq 0.75$	955	248	0.535		0.532	946	251	0.540		0.535
$t \leq 0.5$	960	223	0.536		0.531	973	272	0.536		0.530
$t \leq 0.25$	961	260	0.532		0.517	969	266	0.531		0.528
$0.25 \leq t \leq 0.75$	993	145	0.491		0.466	993	143	0.490		0.481
$t \geq 0.25$	976	128	0.491		0.472	979	126	0.491		0.476
$t \geq 0.5$	995	141	0.474		0.463	991	145	0.478		0.464

The hypothesis that guidance is wasted on the early high-noise steps is confirmed, and more strongly than expected. Guidance restricted to the early steps ($t \geq 0.5$) produces novel QED 0.463–0.464, statistically indistinguishable from the unguided 0.465: it does *nothing*. The entire effect comes from the low-noise steps, which is what the mechanism predicts — at large t the decoded $\mathbf{x}_0$ candidates are mostly unparseable, so they all collapse to the invalid reward and the weights go uniform.

Conversely, dropping the noisiest steps is free or mildly beneficial: $t \leq 0.75$ is the best cell for both N values. Even $t \leq 0.25$, which runs the reward on only the last 8 of 32 steps, retains most of the gain (0.528 at $N = 500$) at a quarter of the reward cost. This is the practical recommendation: restrict guidance to $t \lesssim 0.75$, and use $t \leq 0.25$ if throughput matters.

10 The atomic-measure limitation

The self-normalized estimator replaces $q(\mathbf{x}_0 \mid \mathbf{x}_t)$ with an atomic measure supported on exactly N points, $\hat{q} = \sum_n w_n \delta_{\mathbf{x}_0^{(n)}}$. Under `aggregate.x0` the per-position histogram is

$$\bar{\mathbf{x}}_0^l[k] = \sum_{n=1}^N w_n \mathbf{1}\{\mathbf{x}_0^{(n),l} = k\}, \quad (4)$$

so a token k that appears at position l in *none* of the N candidates receives exactly zero mass. The method is therefore an implicit *stochastic truncation* of the denoiser: since the candidates are drawn i.i.d. from $p_\theta(\mathbf{x}_0^l \mid \mathbf{x}_t)$, a token survives with probability $1 - (1 - p)^N$, so tokens with $p \ll 1/N$ are dropped almost surely. With the 40-token QM9 SMILES vocabulary and $N = 100$, this suppresses everything below roughly the 1% tail.

For uniform diffusion the suppression is severe but not absolute: the posterior in `diffusion.py` contains a term $(1 - \alpha_{ts})(1 - \alpha_s) \pi_{\text{limit}}$ that is positive for every token independently of $\bar{\mathbf{x}}_0$, so unseen tokens retain a small uniform floor — one to two orders of magnitude below an observed token at

these step sizes. For absorbing-state diffusion the posterior is $\bar{\mathbf{x}}_0(\text{move}_t - \text{move}_s)$ with the mask term added separately, and unseen tokens are *exactly* zero.

This is plausibly what caps the achievable property gain, and it cuts both ways: truncating the tail is also the most likely reason validity *rises* under guidance (94–98 % against 97.9 % unguided) while novelty falls, since suppressing low-probability tokens removes both malformed strings and unusual but legitimate ones. The `support_floor` option tests this directly by mixing the denoiser marginal back in,

$$\bar{\mathbf{x}}_0 \leftarrow \frac{\bar{\mathbf{x}}_0 + \epsilon \mathbf{x}_\theta}{1 + \epsilon}, \quad (5)$$

which restores full support for any $\epsilon > 0$ while leaving the tilt intact; $\epsilon = 0$ recovers the plain estimator.

The fix makes things worse. Measured at $N = 300$, $\lambda = 1000$, 1024 samples, the floor destroys the gain rather than extending it:

ϵ	Valid	Novel	QED mean	Novel QED
0 (none)	944	229	0.535	0.526
0.01	965	113	0.508	0.481
0.1	988	120	0.492	0.480
0.5	993	113	0.479	0.474

Even $\epsilon = 0.01$ gives back most of the improvement, and the novel count halves. The reason is that the mixture is applied per position and per step: where the weighted histogram is nearly one-hot, adding $\epsilon \mathbf{x}_\theta$ injects an ϵ chance of an off-distribution token at each of the 32 positions on each of the 32 steps, so on the order of $\epsilon \cdot 32 \cdot 32 \approx 10$ tokens get resampled from the untilted model over a trajectory. That is enough to walk the sample back toward p_θ .

The conclusion is the opposite of the initial worry: the atomic truncation is not a defect limiting the method, it is part of why the method works. Restricting the reverse step to tokens that at least one high-reward candidate proposed acts like nucleus sampling with a $1/N$ threshold, which is also the most plausible explanation for validity holding at 94–97 % under guidance. Support should be widened by raising N — which draws more atoms from the correct tilted region — not by blending in the untilted marginal.

A note on combining with D-CBG. The two mechanisms act on different objects and look complementary. D-CBG reshapes the *proposal* through per-token classifier gradients, and our method reweights *whole-sequence* candidates with an exact reward. Drawing the N candidates from a D-CBG-tilted proposal $\tilde{p}_\theta(\mathbf{x}_0 | \mathbf{x}_t)$ and then reweighting with $w_n \propto \exp(\lambda r(\mathbf{x}_0^{(n)})) p_\theta(\mathbf{x}_0^{(n)}) / \tilde{p}_\theta(\mathbf{x}_0^{(n)})$ — both densities being explicit, so the correction is computable — would attack exactly the limitation above, because the atoms would land where the reward is already high instead of where the untilted model happens to sample. Not implemented; recorded here as a direction.

11 Pending runs

The UDLM axis above is complete at both 32 and 150 steps (Section 7), with one gap noted there: no $\lambda = 0$ control at 150 steps. The MDLM (absorbing-state) axis is in progress: both prerequisites finished training for 25k steps — an MDLM generative model (DiT-small, since no MDLM-QM9

checkpoint is published; the released MDLM models are all OpenWebText with a 1024-token GPT-2 vocabulary) and an absorbing-state D-CBG classifier. The comparison sweep is running.

Early MDLM numbers at 1024 samples: unguided gives novel QED 0.461 with only 64.8% validity, and our method at $N = 100$ reaches 0.499–0.500. The low validity reflects that this MDLM is a from-scratch 25k-step model rather than a released checkpoint, so it is a weaker base than the HuggingFace UDL; its novelty is correspondingly much higher (47% of valid against 12% for UDL), meaning it memorizes far less. Cross-model comparisons of absolute numbers are therefore not meaningful — only the within-model deltas are.

MDLM remains the sharper test of Section 10: its posterior has no uniform floor, so tokens no candidate proposed are *exactly* zero rather than merely suppressed, and `aggregate_x0` coincides exactly with `marginal`. The sweep includes the same `support_floor` ablation for that reason.

The clearest remaining gap is **exact** on MDLM. The `mixture_sampling=exact` runs reported in Table 4 are UDL only, and `scripts/server_run_comparison_mdml.sh` fixes `aggregate_x0`; since **exact** on masked diffusion is precisely EDLM’s Algorithm 1 with the energy swapped for a reward, that cell is the direct test of the substitution in the setting EDLM itself used. Note that `aggregate_x0` and `marginal` coincide there, so MDLM has only two distinct variants and **exact** is the one that matters. The λ that worked on UDL will not transfer — the tilt has to be recalibrated against the weaker base model’s reward scale, watching ESS. Regenerate the combined table with

```
rsync -av wp03052@143.248.84.179:~/Synthetic-Data/DLRT/results/ ./results/
python scripts/make_results_table.py results pdfs
```

which writes `pdfs/sweep_table.tex` and `pdfs/sweep_table.md`.

12 Reproducing

our method

```
python guidance_eval/our_qm9_eval.py \
  data=qm9 data.label_col=qed \
  backbone=hf_dit model=hf model.length=32 \
  model.pretrained_model_name_or_path=kuleshov-group/udl-qm9 \
  diffusion=uniform parameterization=d3pm \
  time_conditioning=True zero_recon_loss=True \
  guidance=ours guidance.reward=qed guidance.lambda_=50 \
  guidance.mixture_sampling=aggregate_x0 guidance.num_x0_samples=30 \
  guidance.t_min=0.0 guidance.t_max=0.5 \
  sampling.steps=32 sampling.num_sample_batches=8 sampling.batch_size=64 \
  sampling.use_cache=False eval.disable_ema=True
```

full sweep + D-CBG baseline (inside tmux)

```
bash scripts/server_train_cbg_classifier.sh uniform qed
bash scripts/server_run_comparison.sh
```

Implementation lives in `our_guidance.py` and `configs/guidance/ours.yaml`; nothing in the original repository was modified.